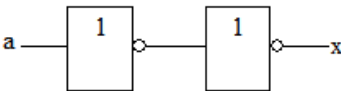
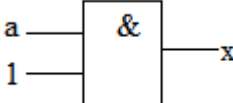
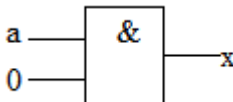
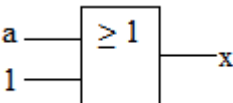
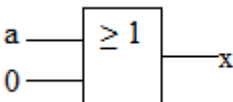
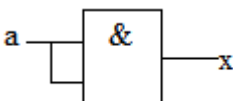
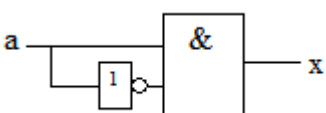
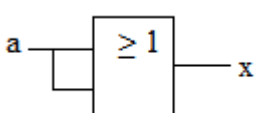
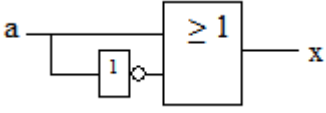


DIGITALTECHNIK

REGELN DER DIGITALTECHNIK/ BOOLESCHE ALGEBRA

Wie auch in der Mathematik, gibt es in der Digitaltechnik Regeln, die **boolesche Algebra**. Durch diese dürfen bestimmte Gatter ersetzt und Schaltfunktionen umgeformt werden.

Zu Beginn geht es um die „einfachen“ Regeln. Suchen Sie sich einen Arbeitspartner und vervollständigen Sie die Wahrheitstabellen, als Beweis, dass die Regeln stimmen.

Nr.	Gesetz/ Funktion	Schaltung	Wahrheitstabelle									
a	$x = \overline{\overline{a}} = a$		<table><tr><th>a</th><th>x</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr></table>	a	x	0	0	1	1			
a	x											
0	0											
1	1											
b	$x = a \wedge 1 = a$		<table><tr><th>$b = 1$</th><th>a</th><th>x</th></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	$b = 1$	a	x	1	0	0	1	1	1
$b = 1$	a	x										
1	0	0										
1	1	1										
c	$x = a \wedge 0 = 0$		<table><tr><th>$b = 0$</th><th>a</th><th>x</th></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	$b = 0$	a	x						
$b = 0$	a	x										
d	$x = a \vee 1 = 1$		<table><tr><th>$b = 1$</th><th>a</th><th>x</th></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	$b = 1$	a	x						
$b = 1$	a	x										
e	$x = a \vee 0 = a$		<table><tr><th>$b = 0$</th><th>a</th><th>x</th></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	$b = 0$	a	x						
$b = 0$	a	x										
f	$x = a \wedge a = a$		<table><tr><th>$b = a$</th><th>a</th><th>x</th></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	$b = a$	a	x						
$b = a$	a	x										
g	$x = a \wedge \bar{a} = 0$		<table><tr><th>$b = \bar{a}$</th><th>a</th><th>x</th></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	$b = \bar{a}$	a	x						
$b = \bar{a}$	a	x										
h	$x = a \vee a = a$		<table><tr><th>$b = a$</th><th>a</th><th>x</th></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	$b = a$	a	x						
$b = a$	a	x										
i	$x = a \vee \bar{a} = 1$		<table><tr><th>$b = \bar{a}$</th><th>a</th><th>x</th></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	$b = \bar{a}$	a	x						
$b = \bar{a}$	a	x										